LAB REPORT

RIP and EIGRP Redistribution with VLANs, DHCP, DNS and Web Server

OBJECTIVES

1. Configure two routers, R1_RIP and R2_EIGRP, using RIP Version 2 and EIGRP AS 100 in different routing domains.
2. Implement two-way route redistribution between RIP and EIGRP for full communication between both networks.
3. Configure VLANs 10, 20 and 30 on Layer 2 switches for traffic separation.
4. Configure router-on-a-stick using 802.1Q trunking so that multiple VLANs can communicate through one router interface.
5. Deploy DHCP servers and use ip helper-address for dynamic IP assignment across VLANs.
6. Configure DNS and HTTP services and verify end-to-end connectivity using domain name and web access.
7. Observe routing behaviour and verify successful redistribution of routes between the two protocols.

COMPONENTS REQUIRED

Device	Qty	Hostname	Role
Router-PT (Gigabit)	2	R1_RIP, R2_EIGRP	RIP and EIGRP routers
Cisco 2960-24TT	2	SW1_RIP, SW2_EIGRP	Layer 2 VLAN switches
Server-PT	4	DHCP1, DHCP2, DNS, Web	DHCP, DNS and HTTP services
PC-PT	4	PC1_VLAN10, PC2_VLAN20, PC3_VLAN10, PC4_VLAN20	End-user hosts

Protocols Used

- RIP Version 2
- EIGRP AS 100
- Route Redistribution
- IEEE 802.1Q VLAN Trunking
- DHCP with IP Helper-Address
- DNS and HTTP

THEORY

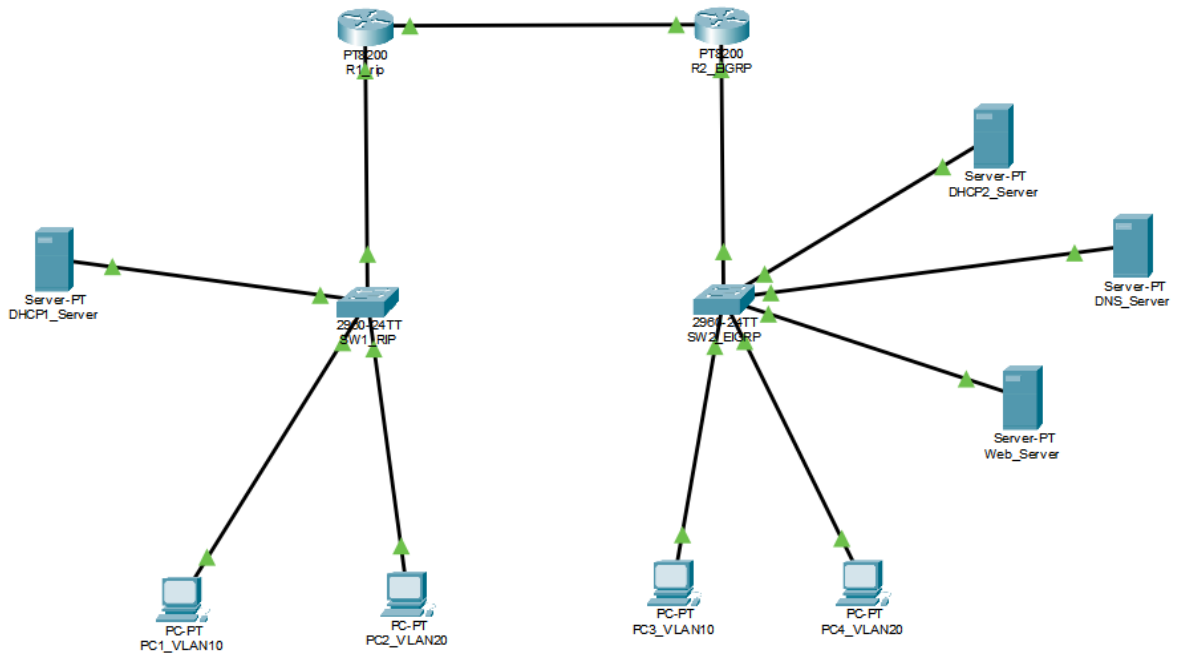
RIP Version 2 is a classless distance-vector routing protocol that uses hop count as its metric. It supports VLSM and CIDR and is suitable for small networks. EIGRP is an advanced Cisco routing protocol that uses the DUAL algorithm and provides fast convergence. It uses bandwidth and delay in its composite metric and supports efficient route updates.

In this lab, RIP is used on the left network (R1_RIP side) and EIGRP is used on the right network (R2_EIGRP side). Since both routing protocols are different, route redistribution is required so that networks learned by RIP can be shared with EIGRP and networks learned by EIGRP can be shared with RIP.

VLANs are used to divide a switch into multiple logical networks. Because several VLANs exist on each side but each router has only one LAN physical interface, router-on-a-stick is used. In this method, one physical router interface is divided into multiple subinterfaces, and each subinterface acts as the default gateway for one VLAN using 802.1Q encapsulation.

DHCP is used to assign IP addresses dynamically to hosts. Since the DHCP server is located in a different VLAN from some clients, ip helper-address is used on router subinterfaces to forward DHCP broadcast requests to the correct DHCP server. DNS resolves domain names to IP addresses, and the web server tests application-level connectivity.

NETWORK DESIGN



VLAN Plan

Side	VLAN	Network	Purpose
RIP Side	VLAN 10	192.168.10.0/24	Users
RIP Side	VLAN 20	192.168.20.0/24	Users
RIP Side	VLAN 30	192.168.30.0/24	Servers
EIGRP Side	VLAN 10	172.16.10.0/24	Users
EIGRP Side	VLAN 20	172.16.20.0/24	Users
EIGRP Side	VLAN 30	172.16.30.0/24	Servers
WAN Link	---	10.10.10.0/30	Router to Router

IP ADDRESSING SCHEME

Device	Interface	IP Address	Subnet Mask	Gateway
R1_RIP	Gig0/0/0.10	192.168.10.1	255.255.255.0	---
R1_RIP	Gig0/0/0.20	192.168.20.1	255.255.255.0	---
R1_RIP	Gig0/0/0.30	192.168.30.1	255.255.255.0	---
R1_RIP	Gig0/0/1	10.10.10.1	255.255.255.252	---
R2_EIGRP	Gig0/0/0.10	172.16.10.1	255.255.255.0	---
R2_EIGRP	Gig0/0/0.20	172.16.20.1	255.255.255.0	---
R2_EIGRP	Gig0/0/0.30	172.16.30.1	255.255.255.0	---
R2_EIGRP	Gig0/0/1	10.10.10.2	255.255.255.252	---
DHCP1_Server	Fa0	192.168.30.10	255.255.255.0	192.168.30.1
DHCP2_Server	Fa0	172.16.30.10	255.255.255.0	172.16.30.1
DNS_Server	Fa0	172.16.30.20	255.255.255.0	172.16.30.1
Web_Server	Fa0	172.16.30.30	255.255.255.0	172.16.30.1
PC1_VLAN10	Fa0	DHCP	---	192.168.10.1
PC2_VLAN20	Fa0	DHCP	---	192.168.20.1
PC3_VLAN10	Fa0	DHCP	---	172.16.10.1
PC4_VLAN20	Fa0	DHCP	---	172.16.20.1

PROCEDURE

Step 1: Configure SW1_RIP

```
enable
configure terminal
hostname SW1_RIP

vlan 10
name VLAN10_USERS
exit

vlan 20
name VLAN20_USERS
exit
```

```
vlan 30
name VLAN30_SERVERS
exit

interface fastEthernet 0/1
description Trunk_to_R1
switchport mode trunk
exit

interface fastEthernet 0/2
description DHCP1_Server
switchport mode access
switchport access vlan 30
exit

interface fastEthernet 0/3
description PC1_VLAN10
switchport mode access
switchport access vlan 10
exit

interface fastEthernet 0/4
description PC2_VLAN20
switchport mode access
switchport access vlan 20
exit

interface vlan 10
ip address 192.168.10.2 255.255.255.0
no shutdown
exit

ip default-gateway 192.168.10.1
end
write memory
```

Step 2: Configure SW2_EIGRP

```
enable
configure terminal
hostname SW2_EIGRP
```

```
vlan 10
name VLAN10_USERS
exit

vlan 20
name VLAN20_USERS
exit

vlan 30
name VLAN30_SERVERS
exit

interface fastEthernet 0/1
description Trunk_to_R2
switchport mode trunk
exit

interface fastEthernet 0/2
description DHCP2_Server
switchport mode access
switchport access vlan 30
exit

interface fastEthernet 0/3
description DNS_Server
switchport mode access
switchport access vlan 30
exit

interface fastEthernet 0/4
description Web_Server
switchport mode access
switchport access vlan 30
exit

interface fastEthernet 0/5
description PC3_VLAN10
switchport mode access
switchport access vlan 10
exit

interface fastEthernet 0/6
description PC4_VLAN20
```

```
switchport mode access
switchport access vlan 20
exit

interface vlan 10
ip address 172.16.10.2 255.255.255.0
no shutdown
exit

ip default-gateway 172.16.10.1
end
write memory
```

Step 3: Configure R1_RIP

```
enable
configure terminal
hostname R1_RIP

interface GigabitEthernet0/0/0
no ip address
no shutdown
exit

interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
ip helper-address 192.168.30.10
exit

interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.30.10
exit

interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit

interface GigabitEthernet0/0/1
ip address 10.10.10.1 255.255.255.252
```

```
no shutdown
exit

router rip
version 2
network 192.168.10.0
network 192.168.20.0
network 192.168.30.0
network 10.10.10.0
no auto-summary
redistribute eigrp 100 metric 5
exit

router eigrp 100
network 10.10.10.0 0.0.0.3
no auto-summary
redistribute rip metric 10000 100 255 1 1500
exit

end
write memory
```

Step 4: Configure R2_EIGRP

Note: Router uses GigabitEthernet0/0/0 (LAN) and GigabitEthernet0/0/1 (WAN).

```
enable
configure terminal
hostname R2_EIGRP

interface GigabitEthernet0/0/0
no ip address
no shutdown
exit

interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 172.16.10.1 255.255.255.0
ip helper-address 172.16.30.10
exit

interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 172.16.20.1 255.255.255.0
```

```
ip helper-address 172.16.30.10
exit

interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 172.16.30.1 255.255.255.0
exit

interface GigabitEthernet0/0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
exit

router eigrp 100
network 172.16.10.0 0.0.0.255
network 172.16.20.0 0.0.0.255
network 172.16.30.0 0.0.0.255
network 10.10.10.0 0.0.0.3
no auto-summary
redistribute rip metric 10000 100 255 1 1500
exit

router rip
version 2
network 10.10.10.0
no auto-summary
redistribute eigrp 100 metric 5
exit

end
write memory
```

Step 5: Configure DHCP1_Server

Desktop > IP Configuration

Field	Value
IP Address	192.168.30.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	172.16.30.20

Services > DHCP > ON — Delete default pool, then create:

Pool Name	Default Gateway	Start IP
VLAN10	192.168.10.1	192.168.10.50
VLAN20	192.168.20.1	192.168.20.50

Step 6: Configure DHCP2_Server

Desktop > IP Configuration

Field	Value
IP Address	172.16.30.10
Subnet Mask	255.255.255.0
Default Gateway	172.16.30.1
DNS Server	172.16.30.20

Services > DHCP > ON — Delete default pool, then create:

Pool Name	Default Gateway	Start IP
VLAN10	172.16.10.1	172.16.10.50
VLAN20	172.16.20.1	172.16.20.50

Step 7: Configure DNS_Server

Desktop > IP Configuration

Field	Value
IP Address	172.16.30.20
Subnet Mask	255.255.255.0
Default Gateway	172.16.30.1
DNS Server	172.16.30.20

Services > DNS > ON — Add records:

Name	Address
www.networklab.local	172.16.30.30
networklab.local	172.16.30.30

Step 8: Configure Web_Server

Desktop > IP Configuration

Field	Value
IP Address	172.16.30.30
Subnet Mask	255.255.255.0
Default Gateway	172.16.30.1
DNS Server	172.16.30.20

Services > HTTP > ON — Edit index.html:

```
<html>
<head><title>Network Lab</title></head>
<body>
<h1>Welcome to RIP and EIGRP Redistribution Lab done by utsav bhatta.</h1>
<p>DNS name: www.networklab.local</p>
</body>
</html>
```

Step 9: Configure PCs

Click each PC > Desktop > IP Configuration > select DHCP

PC	Expected IP After DHCP
PC1_VLAN10	192.168.10.50
PC2_VLAN20	192.168.20.50
PC3_VLAN10	172.16.10.50
PC4_VLAN20	172.16.20.50

VERIFICATION

On R1_RIP — type end first, then:

show ip route
show ip protocols
show ip eigrp neighbors

```
*Mar 21, 13:44:10.444: %SYS-5-CONFIG_I: Configured from console by console
R1_RIP#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.10.1/32 is directly connected, GigabitEthernet0/0/1
     172.16.0.0/24 is subnetted, 3 subnets
D       172.16.10.0 [90/28416] via 10.10.10.2, 00:25:04, GigabitEthernet0/0/1
D       172.16.20.0 [90/28416] via 10.10.10.2, 00:25:04, GigabitEthernet0/0/1
D       172.16.30.0 [90/28416] via 10.10.10.2, 00:25:04, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0/0.30

R1_RIP#
```

```
R1_RIP#show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 100, rip
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
     10.10.10.0/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.10.2      90            727497
  Distance: internal 90 external 170

Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 22 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, eigrp 100
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0/1  22
  GigabitEthernet0/0/0.1022
  GigabitEthernet0/0/0.2022
  GigabitEthernet0/0/0.3022
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
            192.168.10.0
            192.168.20.0
            192.168.30.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
Distance: (default is 120)
```

```
R1_RIP#
R1_RIP#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.10.10.2      Gig0/0/1       14   00:27:07  40     1000  0   4

R1_RIP#
```

On R2_EIGRP — type end first, then:

show ip route
show ip protocols
show ip eigrp neighbors

```
R2_EIGRP>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.10.2/32 is directly connected, GigabitEthernet0/0/1
     172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       172.16.10.1/32 is directly connected, GigabitEthernet0/0/0.10
C       172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       172.16.20.1/32 is directly connected, GigabitEthernet0/0/0.20
C       172.16.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L       172.16.30.1/32 is directly connected, GigabitEthernet0/0/0.30
R    192.168.10.0/24 [120/1] via 10.10.10.1, 00:00:04, GigabitEthernet0/0/1
R    192.168.20.0/24 [120/1] via 10.10.10.1, 00:00:04, GigabitEthernet0/0/1
R    192.168.30.0/24 [120/1] via 10.10.10.1, 00:00:04, GigabitEthernet0/0/1
```

```
R2_EIGRP>
R2_EIGRP>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 100, rip
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
     172.16.10.0/24
     172.16.20.0/24
     172.16.30.0/24
     10.10.10.0/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.10.1      90            676673
  Distance: internal 90 external 170

Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 7 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, eigrp 100
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0/1  22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
            10.10.10.1           120      00:00:18
Distance: (default is 120)
R2_EIGRP>
```

```
R2_EIGRP>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.10.10.1      Gig0/0/1       13   00:32:17  40     1000  0   13
```

VERIFICATION ON SWITCHES:

On Switches

show vlan brief
show interfaces trunk

SW_RIP:

```
SW1_RIP>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   VLAN10_USERS                     active    Fa0/3
20   VLAN20_USERS                     active    Fa0/4
30   VLAN30_SERVERS                   active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1_RIP>
```

```
SW1_RIP>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
```

SW_EIGRP:

```
SW2_EIGRP>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   VLAN10_USERS                     active    Fa0/5
20   VLAN20_USERS                     active    Fa0/6
30   VLAN30_SERVERS                   active    Fa0/2, Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2_EIGRP>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
```

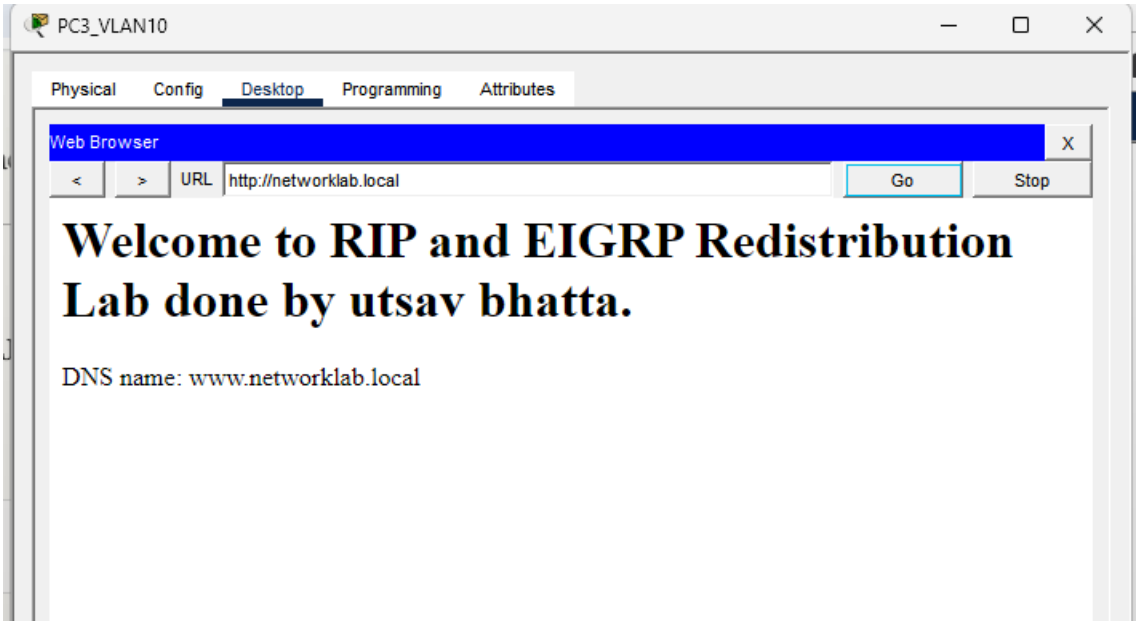
On PCs — Desktop > Command Prompt

ipconfig
ping 192.168.10.1
ping 172.16.30.30

PING TEST IN PC1_VLAN10:

```
C:\>ping 172.16.30.30

Pinging 172.16.30.30 with 32 bytes of data:

Reply from 172.16.30.30: bytes=32 time<1ms TTL=126
Reply from 172.16.30.30: bytes=32 time<1ms TTL=126
Reply from 172.16.30.30: bytes=32 time<1ms TTL=126
Reply from 172.16.30.30: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.30.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Browser test in PC3_VLAN10: Desktop > Web Browser



OBSERVATION

Routing Table on R1_RIP

Route Code	Network	Learned From
C	192.168.10.0/24	Directly connected
C	192.168.20.0/24	Directly connected
C	192.168.30.0/24	Directly connected
C	10.10.10.0/30	Directly connected
D EX	172.16.10.0/24	Redistributed from EIGRP
D EX	172.16.20.0/24	Redistributed from EIGRP
D EX	172.16.30.0/24	Redistributed from EIGRP

Routing Table on R2_EIGRP

Route Code	Network	Learned From
C	172.16.10.0/24	Directly connected
C	172.16.20.0/24	Directly connected
C	172.16.30.0/24	Directly connected
C	10.10.10.0/30	Directly connected
R	192.168.10.0/24	Redistributed from RIP
R	192.168.20.0/24	Redistributed from RIP
R	192.168.30.0/24	Redistributed from RIP

VLAN Assignment on Switches

Switch	Port	VLAN	Device
SW1_RIP	Fa0/1	Trunk	R1_RIP
SW1_RIP	Fa0/2	30	DHCP1_Server
SW1_RIP	Fa0/3	10	PC1_VLAN10
SW1_RIP	Fa0/4	20	PC2_VLAN20
SW2_EIGRP	Fa0/1	Trunk	R2_EIGRP
SW2_EIGRP	Fa0/2	30	DHCP2_Server

Switch	Port	VLAN	Device
SW2_EIGRP	Fa0/3	30	DNS_Server
SW2_EIGRP	Fa0/4	30	Web_Server
SW2_EIGRP	Fa0/5	10	PC3_VLAN10
SW2_EIGRP	Fa0/6	20	PC4_VLAN20

RESULT

The project was completed successfully. RIP and EIGRP networks were connected through two-way redistribution. VLAN communication, DHCP, DNS, and web services worked properly, and devices from both sides communicated successfully.

DISCUSSION

This project showed how different routing protocols can work together using route redistribution. It also demonstrated the use of router-on-a-stick for VLANs and DHCP relay for automatic IP assignment across networks. Proper metrics and correct configuration were necessary for successful communication.

CONCLUSION

The project proved that RIP and EIGRP can be integrated into one working network. With proper VLAN setup, redistribution, and network services, full end-to-end connectivity was achieved.